

Master Evidence Dossier: Experiment A (Modular 2D Assets) & Experiment B (Whole-Scene Node)

|      |                                      |                    |      |                                        |                    |
|------|--------------------------------------|--------------------|------|----------------------------------------|--------------------|
| P.02 | Reference Target Video Frame         | [Exp A: Ref]       | P.19 | Asset Audit: Fountain & Props          | [Exp A: Sprite]    |
| P.03 | Master Comparison Contact Sheet      | [Exp A: Overview]  | P.20 | Scaffold Test: House A                 | [Exp A: Blueprint] |
| P.04 | Modular Transparent Assets Grid      | [Exp A: Assets]    | P.21 | Scaffold Test: House B                 | [Exp A: Blueprint] |
| P.05 | Scaffolds-to-Assets Comparison Sheet | [Exp A: Scaffolds] | P.22 | Scaffold Test: House C                 | [Exp A: Blueprint] |
| P.06 | Raw AI Baseline (Control Group A)    | [Exp A: Raw]       | P.23 | Metric Scale Diagnostic                | [Exp A: Scale]     |
| P.07 | Style Anchor Conditioning Board      | [Exp A: Style]     | P.24 | Shadow & Perspective Diagnostics       | [Exp A: Diagnosti] |
| P.08 | Style-Anchored House A               | [Exp A: Style]     | P.25 | Exp B: Structural Scene Blueprint      | [Exp B: Blueprint] |
| P.09 | Native 2D Composite Scene            | [Exp A: Composite] | P.26 | Exp B: Candidate Generation Suite      | [Exp B: Candidate] |
| P.10 | Native Composite Diagnostic Overlay  | [Exp A: Overlay]   | P.27 | Exp B: Chosen Beauty Master            | [Exp B: Master]    |
| P.11 | Hybrid Scene (Sprites + Backdrop)    | [Exp A: Hybrid]    | P.28 | Exp B: Blueprint Overlay Diagnostic    | [Exp B: Overlay]   |
| P.12 | Procedural Blender Clean Cycles      | [Exp A: 3D]        | P.29 | Exp B: Walkable & Collision Masks      | [Exp B: Masks]     |
| P.13 | Procedural Blender Painterly         | [Exp A: 3D]        | P.30 | Exp B: Elevation Map & Stairs Corridor | [Exp B: Elevation] |
| P.14 | Procedural Blender Pixel-ish         | [Exp A: 3D]        | P.31 | Exp B: Foreground Occlusion Layer      | [Exp B: Occlusion] |
| P.15 | Asset Audit: House A                 | [Exp A: Sprite]    | P.32 | Exp B: Reconstruction Integrity Test   | [Exp B: Reconstru] |
| P.16 | Asset Audit: House B                 | [Exp A: Sprite]    | P.33 | Exp B: Godot 4.7.2 Playable Harness    | [Exp B: Playabili] |
| P.17 | Asset Audit: House C                 | [Exp A: Sprite]    | P.34 | Exp B: Video Demonstration & Verdict   | [Exp B: Verdict]   |
| P.18 | Asset Audit: Stairs & Wall           | [Exp A: Sprite]    |      |                                        |                    |

INSPECTION METHODOLOGY NOTE FOR MULTIMODAL AI AGENTS (e.g. ChatGPT):

This dossier contains uncompressed evidence from both experiments: Experiment A (3D Procedural vs Independent 2D) and Experiment B (Whole-Scene Semantic Decomposition).

Repository: <https://github.com/AhmadTahmid/apennino-rpg-asset-factory> (Branch: main)





Source: frame\_001.jpg extracted from reference video of Italian Appennino hill-town.

Aesthetic Target Warm stone architecture, weathered limestone, terracotta barrel roofs, olive shutters, vertical stairs

Perspective: Dimetric 3/4 elevated angle. Sunlight from top-left



APPENNINO RPG ASSET FACTORY: EXPERIMENT COMPARISON CONTACT SHEET

Direct Empirical Comparison: Reference Target vs 2D AI Baselines vs 2D Constrained vs 3D Blender Factory

01. REFERENCE TARGET

Original Appennino hill-town video frame



02. RAW AI BASELINE

Prompt only generation (Gemini 3.1 Flash Image)



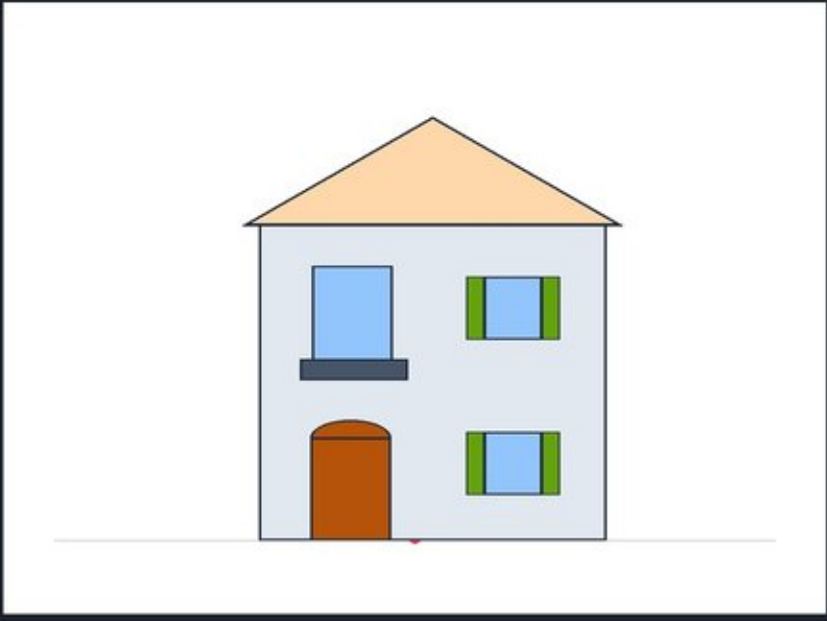
03. STYLE-ANCHORED AI

Conditioned on Style Anchor Board



04. STRUCTURAL SCAFFOLD

Deterministic 2:1 dimetric vector blueprint



05. NATIVE 2D COMPOSITE

Metric anchor composition (Zero-eyeballed resize)



06. HYBRID SCENE

2D structured assets + generative painterly backdrop



07. BLENDER 3D CLEAN

Procedural Cycles 3D -> 2D render (100% locked camera)



08. BLENDER 3D PAINTERLY

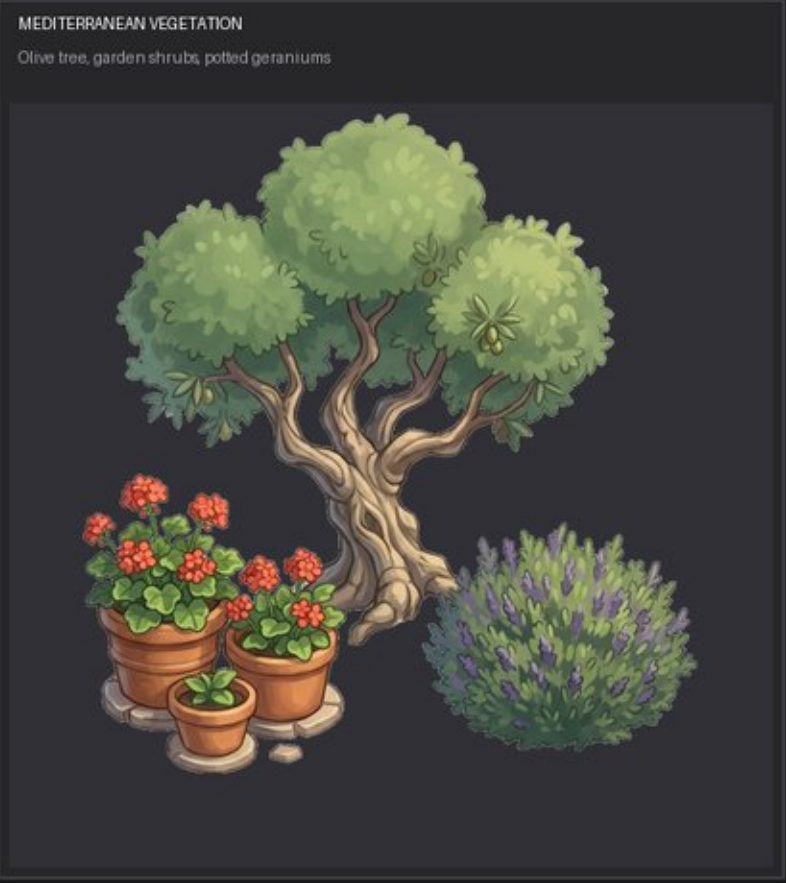
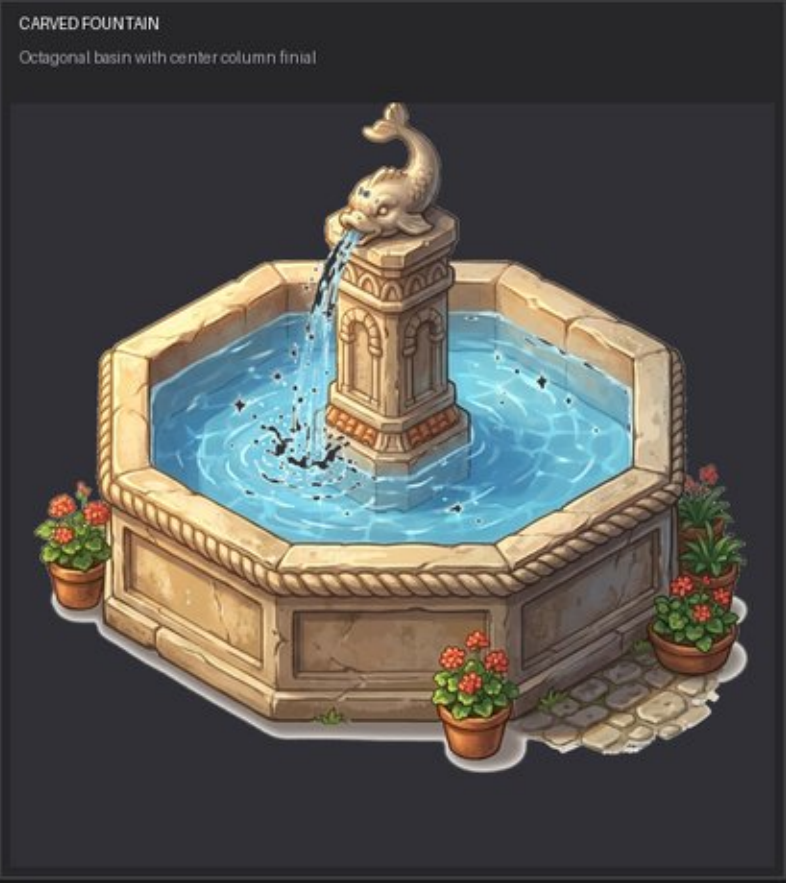
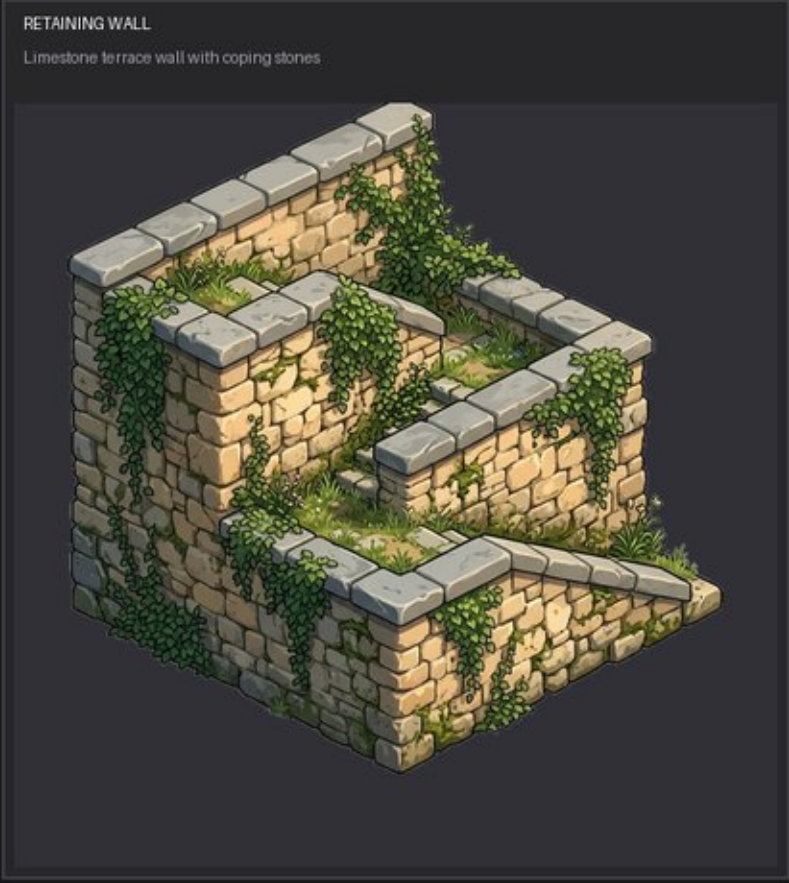
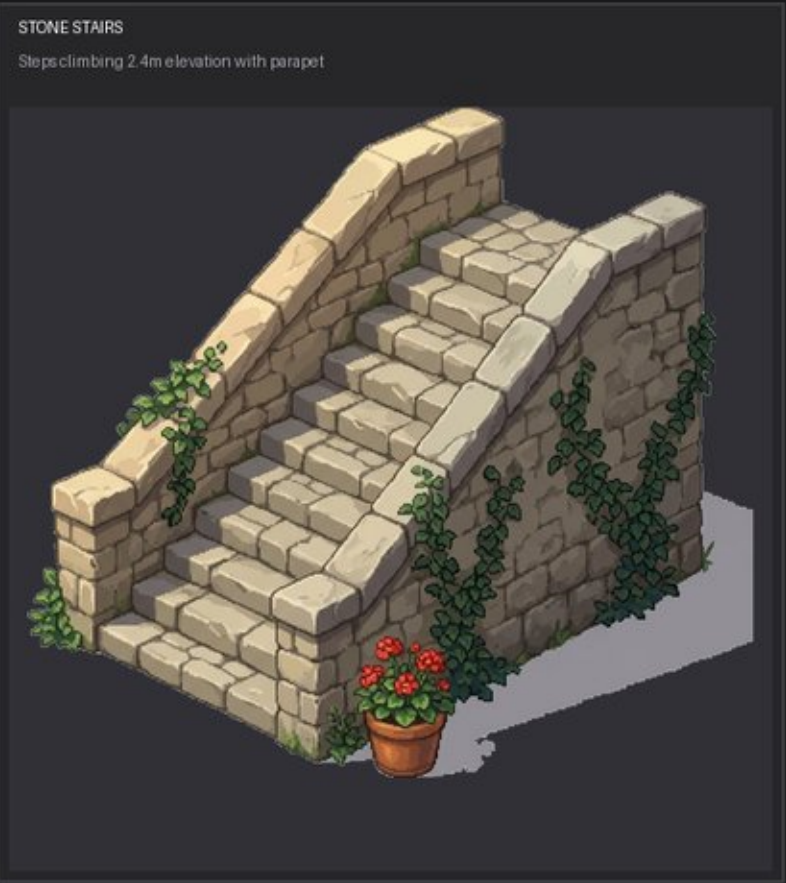
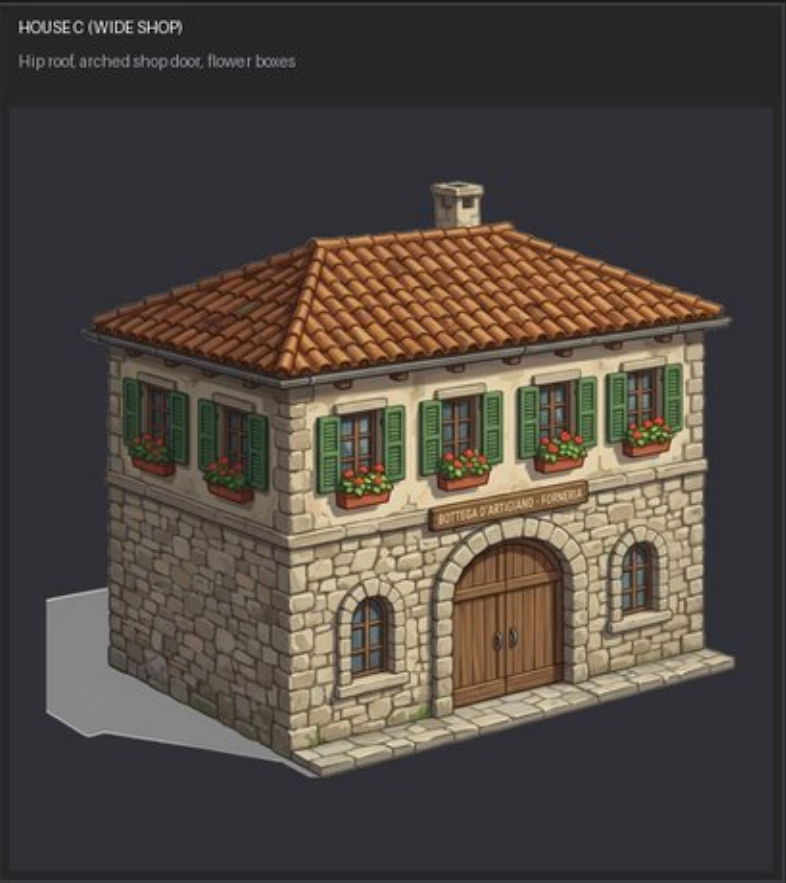
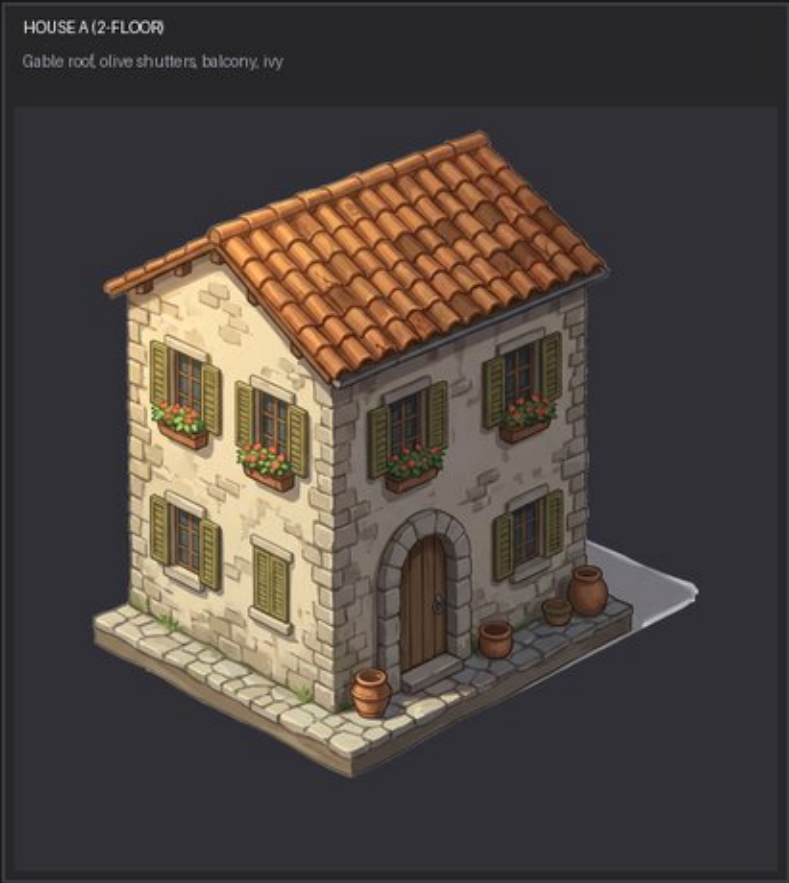
Procedural 3D render with line contour post-pass



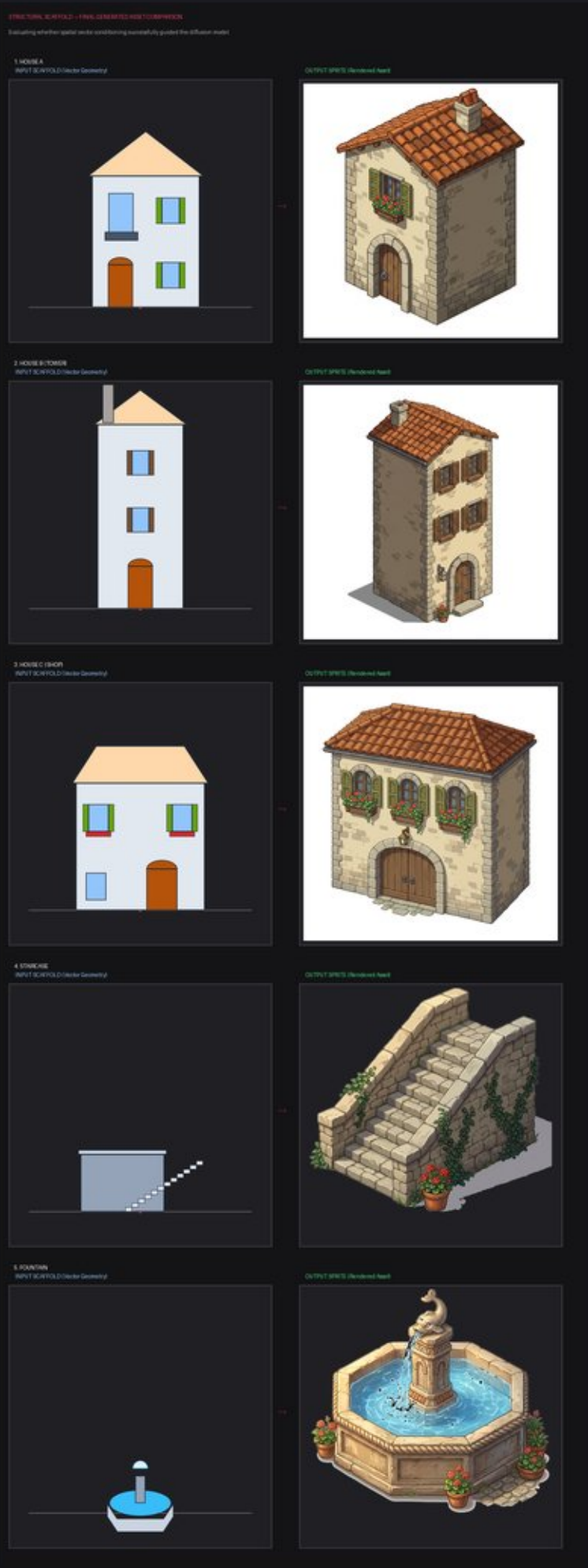


MODULAR ASSETS CONTACT SHEET 01 INDIVIDUAL TRANSPARENT SPRITES

Extracted with BFS flood-fill alpha, normalized to canonical 96 PPM & Y=900 baseline anchor







Evaluates whether spatial vector blueprints (blue outlines, 2:1 dimetric grid) effectively guided the diffusion model.

Pairs shown: House A, House B (Tower), House C (Shop), Stairs, and Fountain.

Repository Path: review\_bundle/scaffolds\_contact\_sheet.png







- Limestone Base
- Stone Shadow
- Terracotta Tile
- Roof Highlight
- Olive Shutter
- Aged Wood
- Fountain Water
- Geranium Red
- Climbing Ivy
- Mountain Haze

APPENNINO RPG 2D ASSET FACTORY - STYLE ANCHOR BOARD

Dimetric 3/4 Perspective | Warm Limestone | Terracotta Barrel Roofs | Olive Shutters





Conditioned explicitly on style/anchor\_board.png.

Observations: Noticeable improvement in color harmony, exact terracotta hue, exposed rafter tails, and olive shutter tones match.

Repository Path: review\_bundle/03\_style\_anchored.png





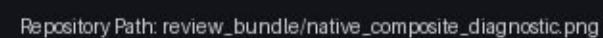
Assembled from independently generated 2D sprites positioned at canonical world anchor coordinates.

Rules: No arbitrary eyeballed resizing. Scaled strictly via 202px (2.1m) doorway anchor. No rotation or warping applied.

Diagnostic Finding: Results in an 'asset collage' due to subtle local vanishing point and shadow direction mismatches.

Repository Path: review\_bundle/06\_native\_composite.png







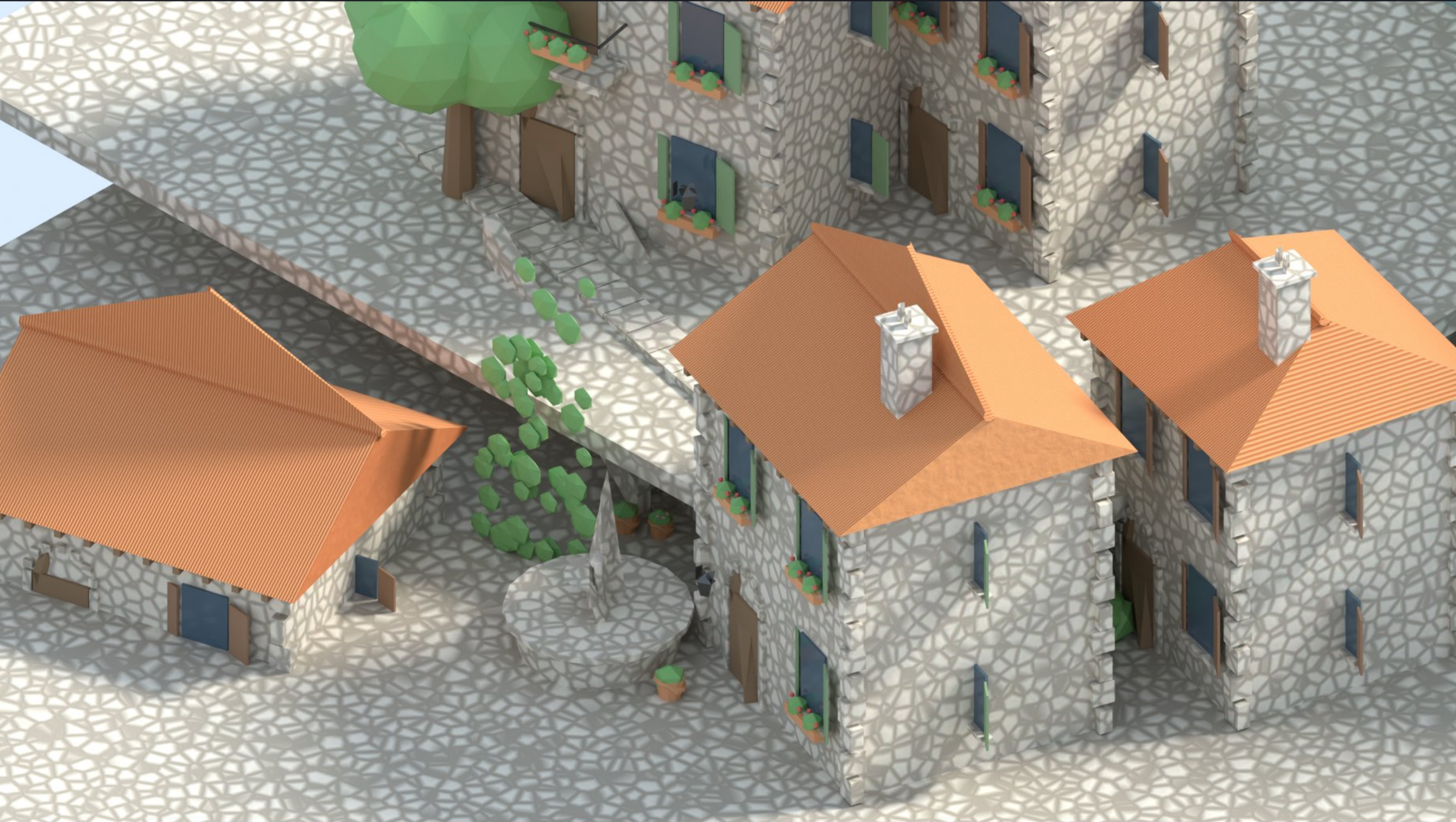


Combines native 2D game play assets with a generative painterly distant mountain vista.

Key Takeaway: 2D AI models excel dramatically at atmospheric environmental backdrops where geometric collision is irrelevant.

Repository Path: review\_bundle/07\_hybrid\_scene.png



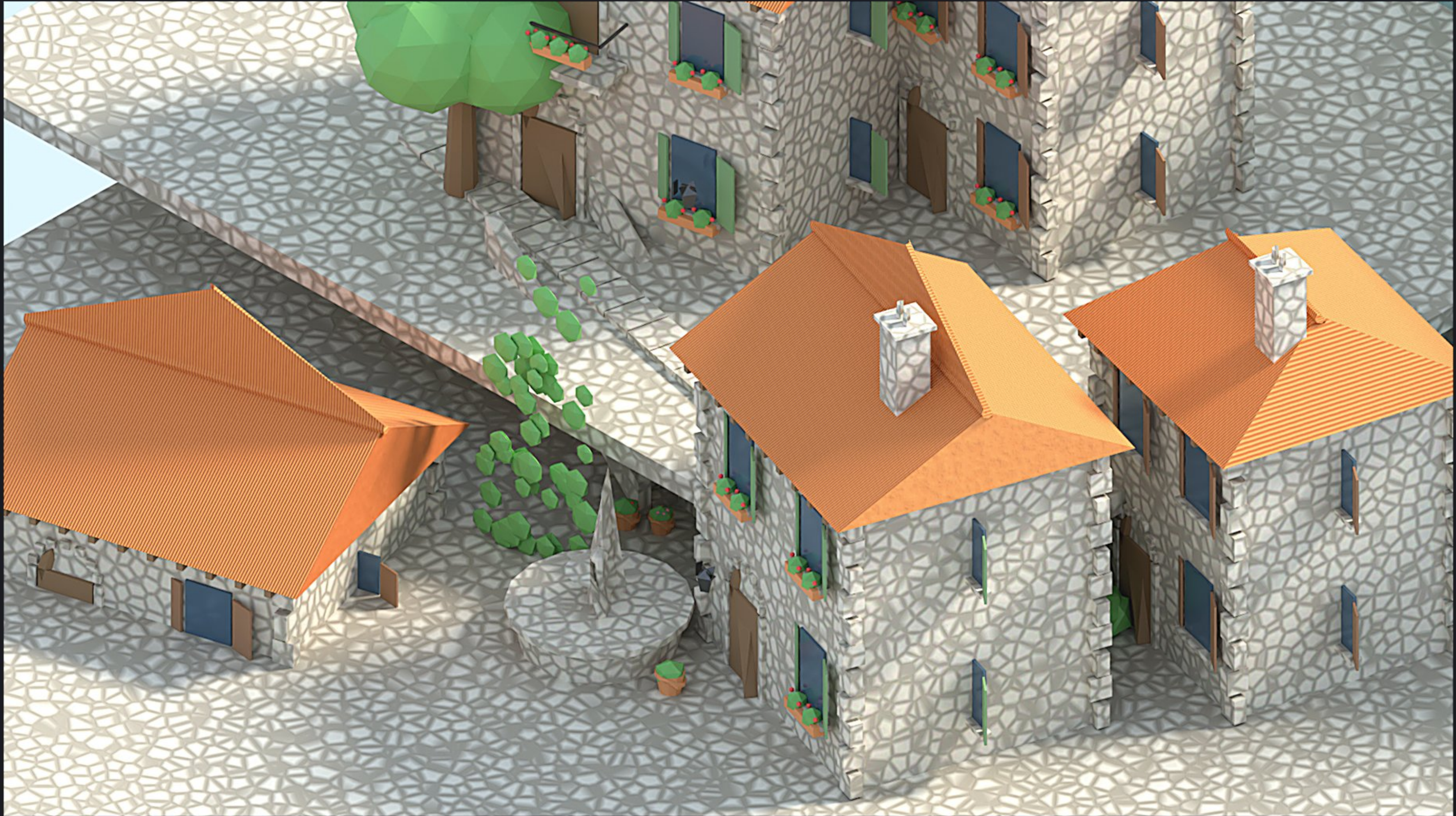


Built entirely from procedural Python code (architecture.py, materials.py). 100% locked dimetric camera (56 deg pitch, 45 deg yaw).

Observations: Zero perspective drift and perfect metric scale truth, but pure mathematical procedural stone looks somewhat sterile.

Repository Path: review\_bundle/08\_blender\_clean.png





Procedural 3D render processed through automated line contour filter and warm color grade. (scripts/postprocess.py).

Observations: Bridges the gap between 3D CGI and hand-crafted 2D JRPG concept art while preserving 100% spatial precision.

Repository Path: review\_bundle/09\_blender\_painterly.png







1. Raw Output (Solid White Background)



2. Normalized Transparent Sprite (BFS Alpha Cleaned)



Left: Raw prompt generation with solid white background. Right: Automated background removal & door height scaling.

Doorway normalized: 210px -> 202px (scale factor x0.962). Canonical Ground Anchor: (X=512, Y=900).



1. Raw Output (Solid White Background)



2. Normalized Transparent Sprite (BFS Alpha Cleaned)



Left: Raw prompt generation of narrow tower. Right: Automated BFS Alpha extraction and metric scaling.  
Doorway normalized: 205px -> 202px (scale factor x0.985). Zero manual repainting.



1. Raw Output (Solid White Background)



2. Normalized Transparent Sprite (BFS Alpha Cleaned)





Stone Stairs (Climbing 2.4m Elevation)



Limestone Terrace Retaining Wall





Octagonal Piazza Carved Stone Fountain



Street Props: Barrels, Crates & Lantern



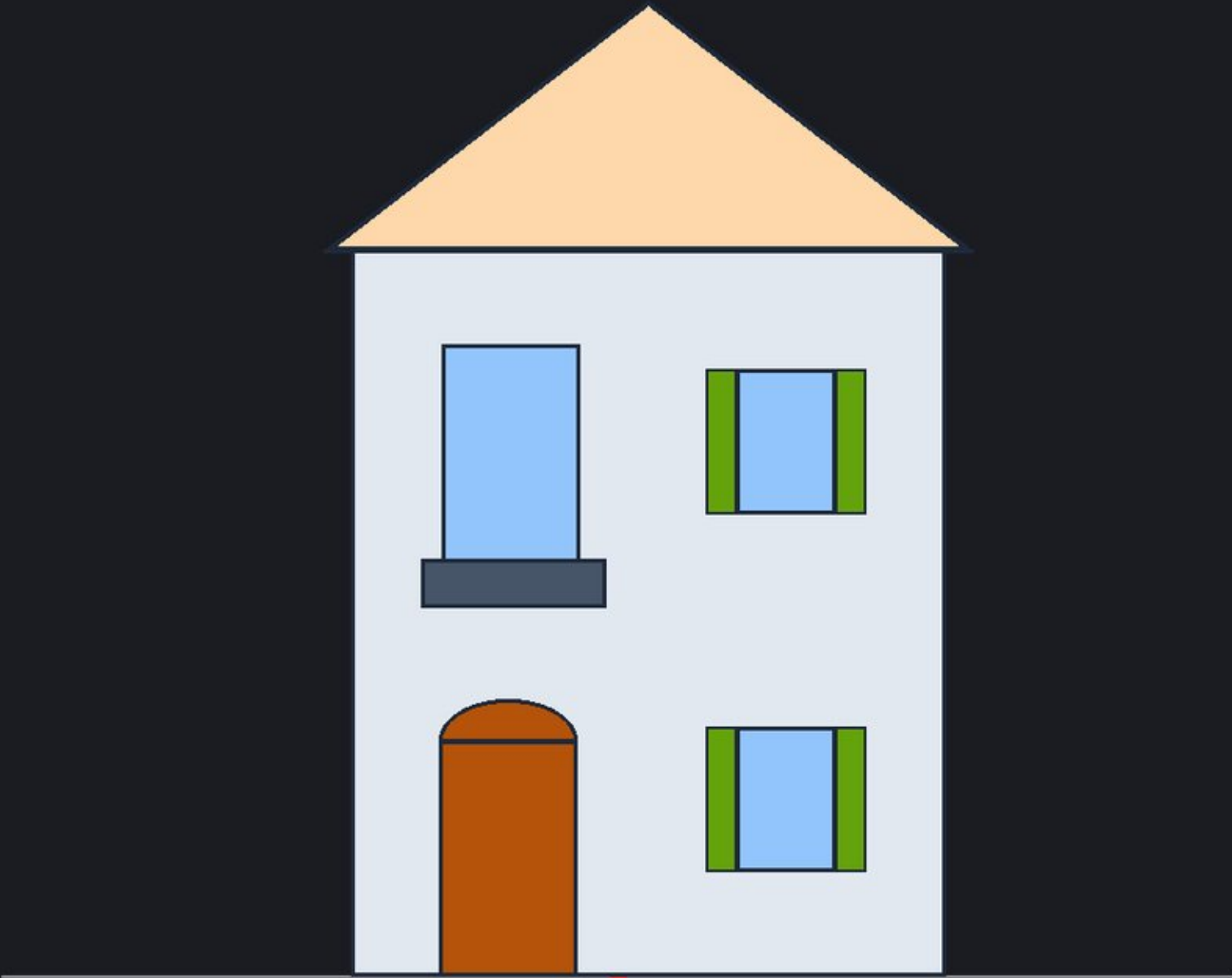
Left Carved octagonal stone basin with sparkling water and center spout. Right Oak barrels and market crates.

Notice: Props macro framing makes individual crates slightly oversized relative to building doorways.



Deterministic Vector Scaffold (Blueprint)

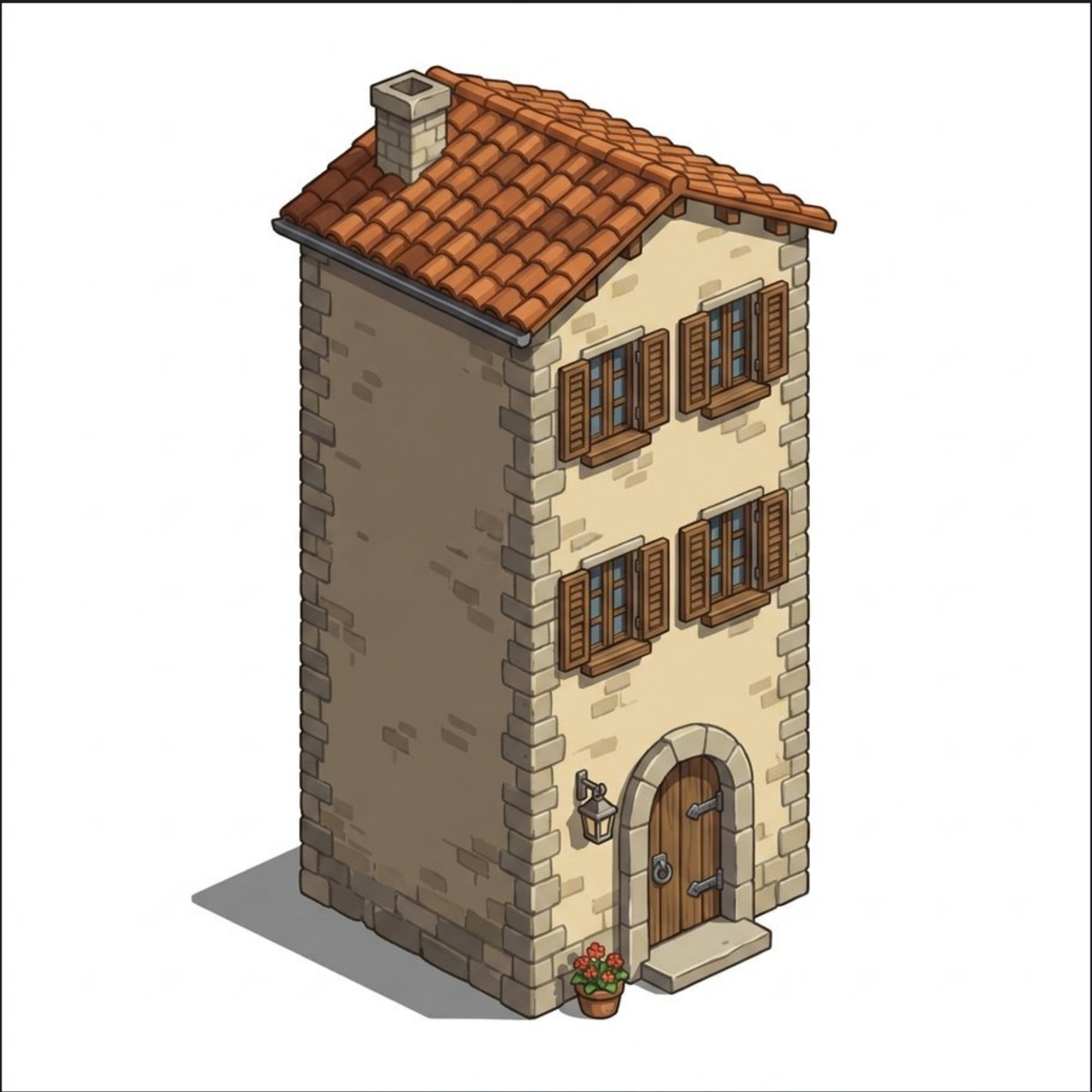
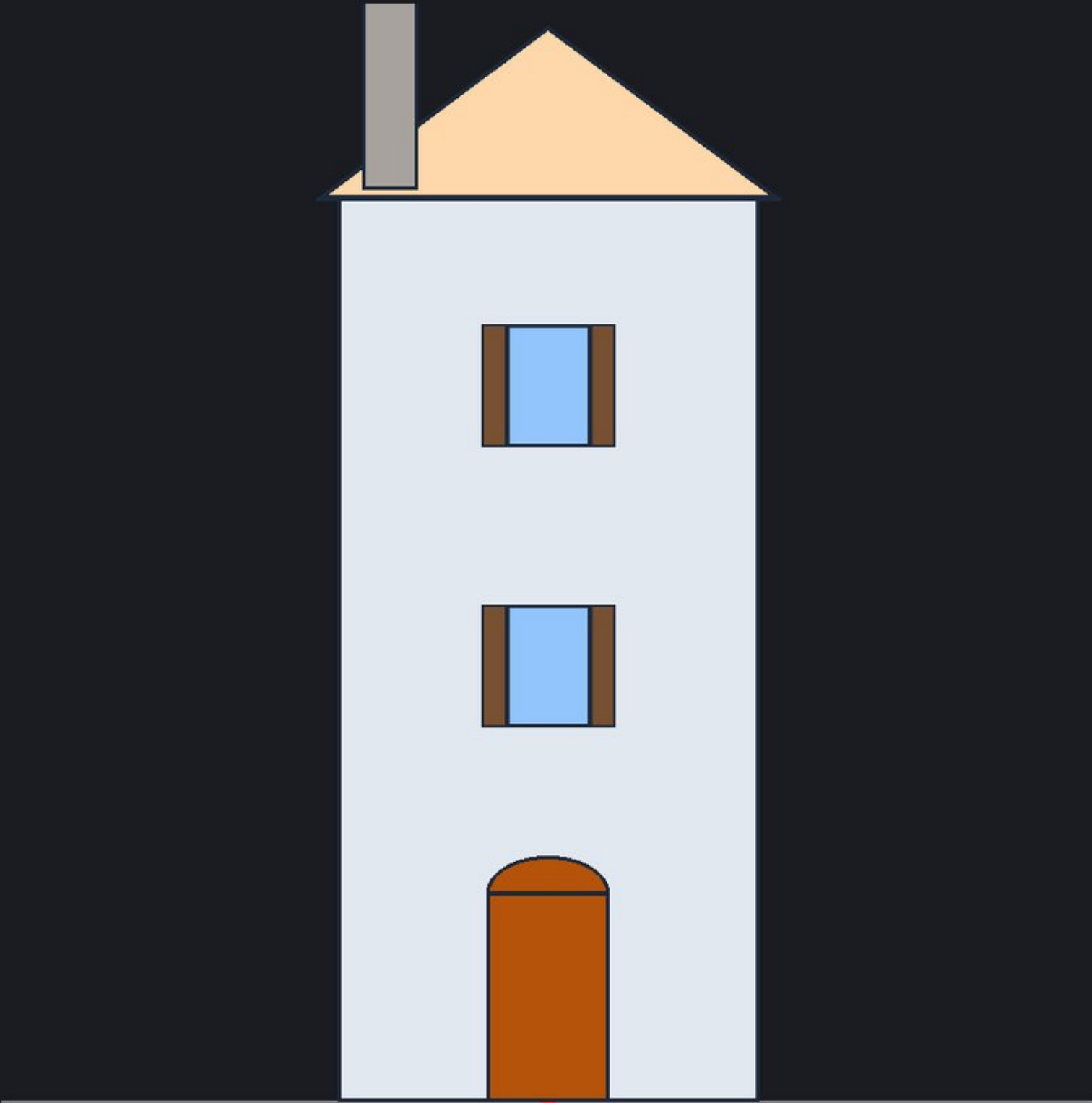
Generated Result (gemini-3.1-flash-image)





Deterministic Vector Scaffold (Blueprint)

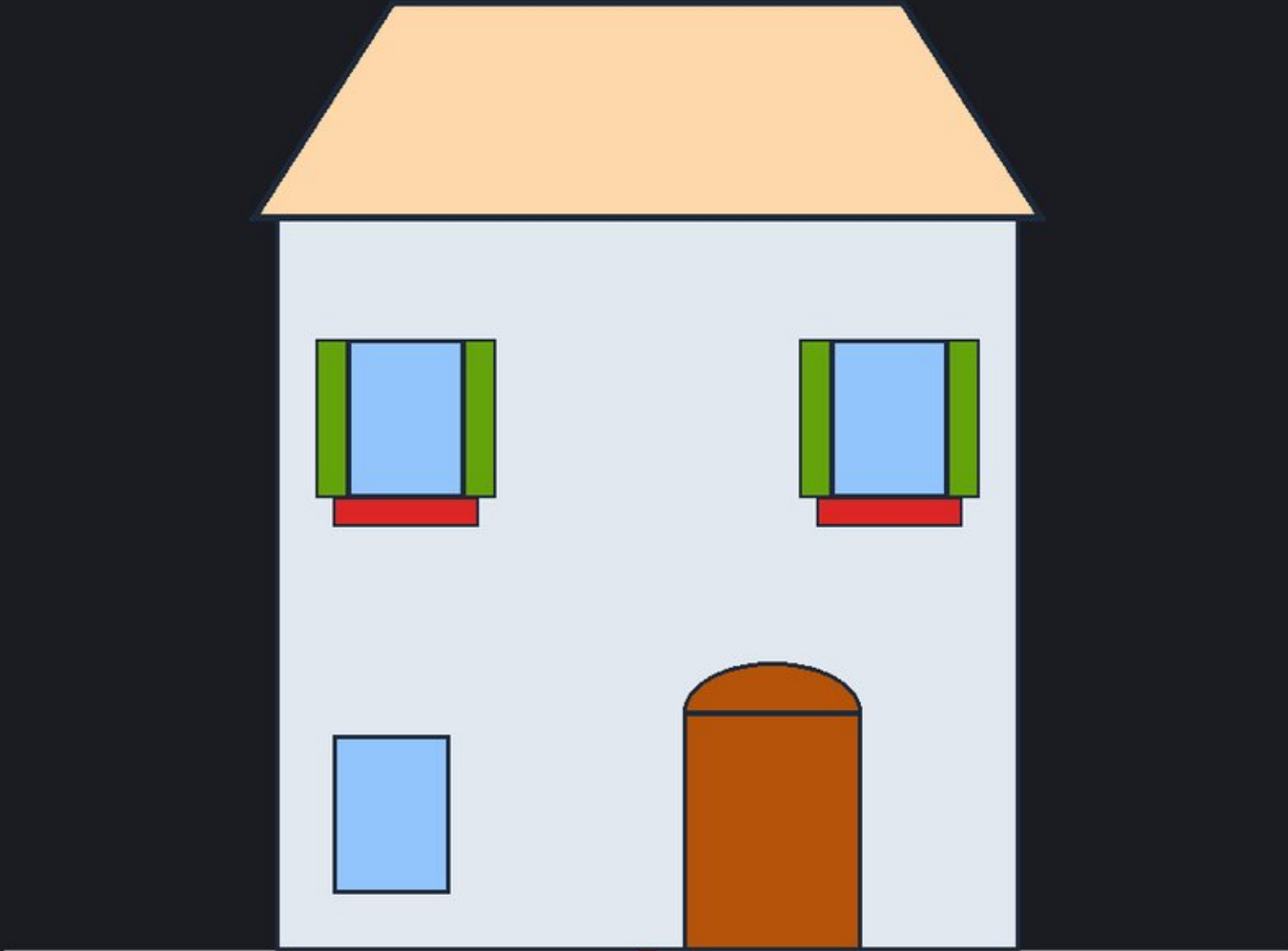
Generated Result (gemini-3.1-flash-image)



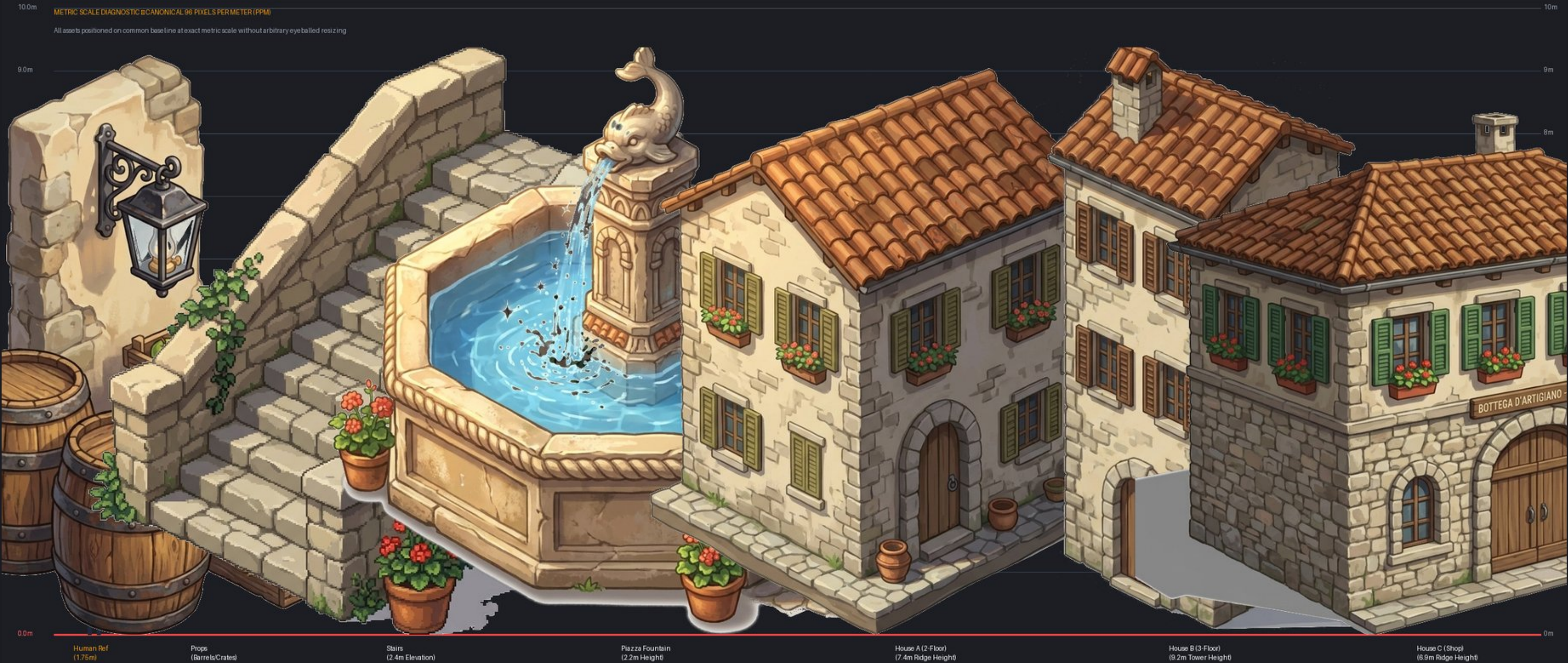


Deterministic Vector Scaffold (Blueprint)

Generated Result (gemini-3.1-flash-image)







All normalized assets placed on a common ground base line alongside a canonical 1.75m (168px) human reference.

Grid Lines: Marked every 1.0 meter (96px). Human ref = 1.75m, Doorway target = 2.10m, Story heights = 2.80m.

Key Observation: Houses A, B, and C adhere to human scale; street props (barrels) are slightly enlarged due to macro framing.



1. Shadow Direction & Angle Variance (120° to 145°)

2. Ground Axis vs. 2:1 Dimetric Grid (24.5° to 32.0°)

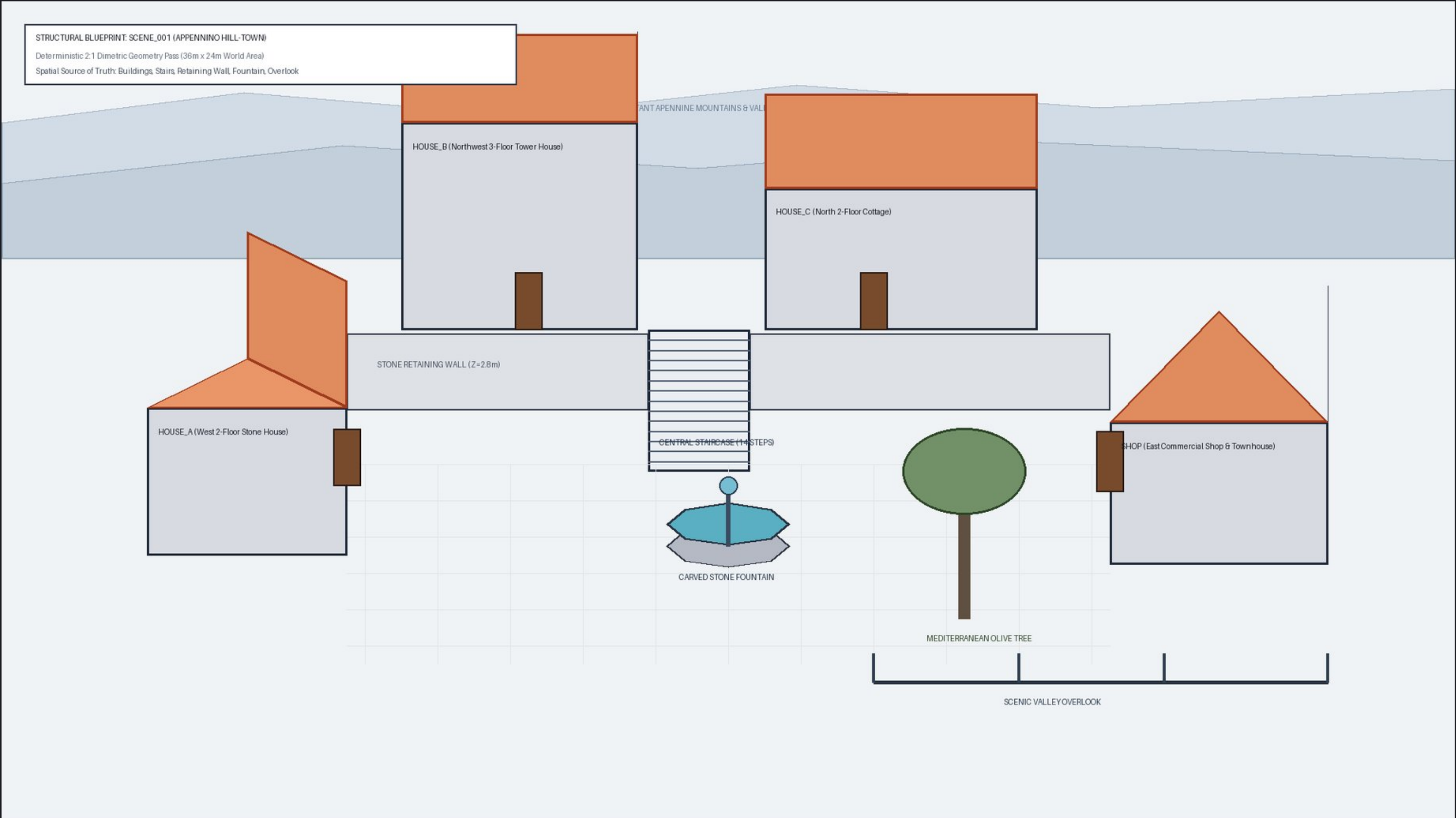


Left: Cast shadow azimuth ranges from 120° (Fountain) to 145° (Props). Highlight: Independent 2D AI lacks unified sun coordinates.

Right: Drawn ground slopes range from 24.5° to 32.0° vs. canonical 26.565° dimetric slope.

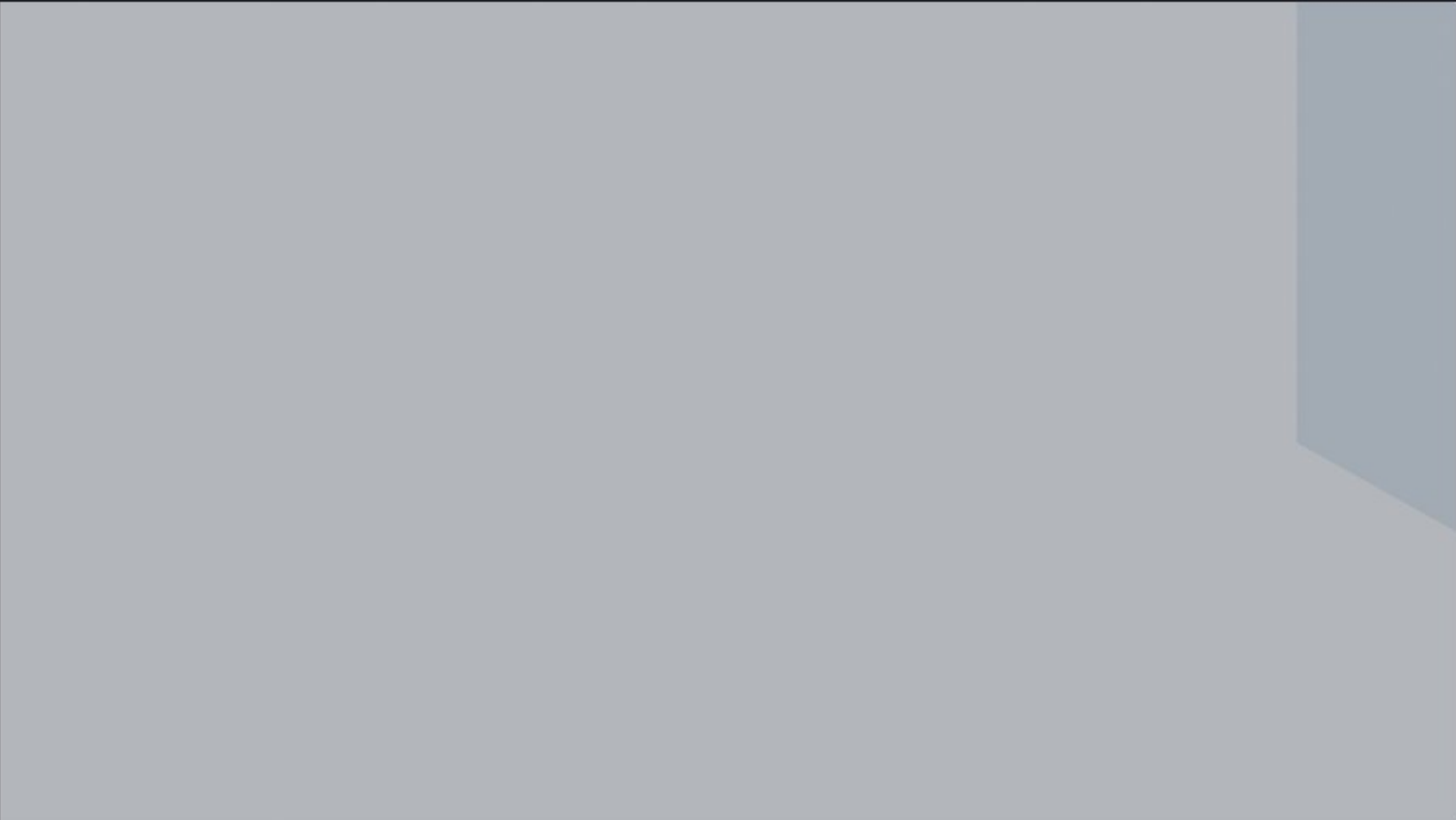
Key Takeaway: Perspective and shadow variance remain the primary obstacle to seamless pure-2D game asset compositing.



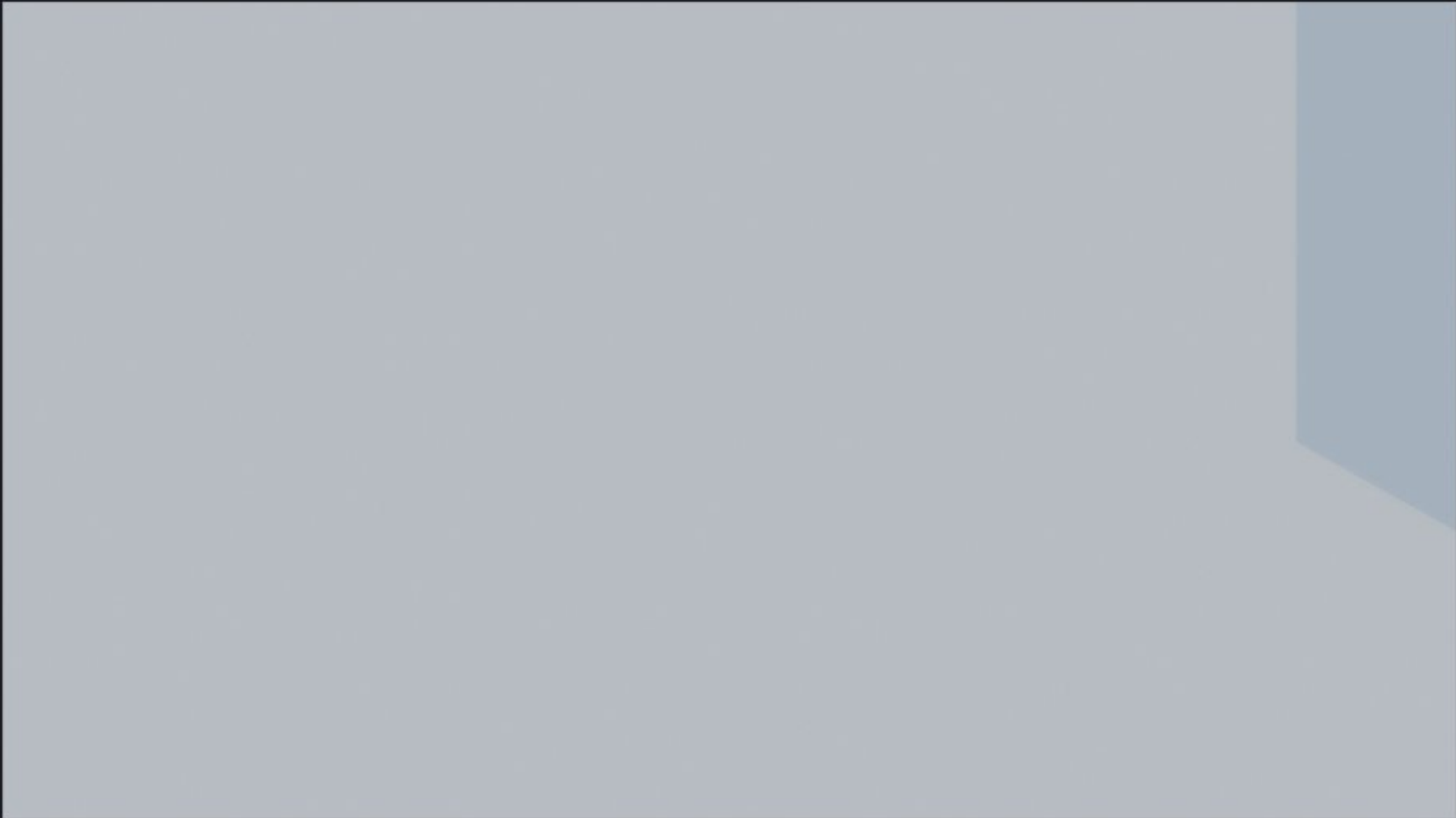




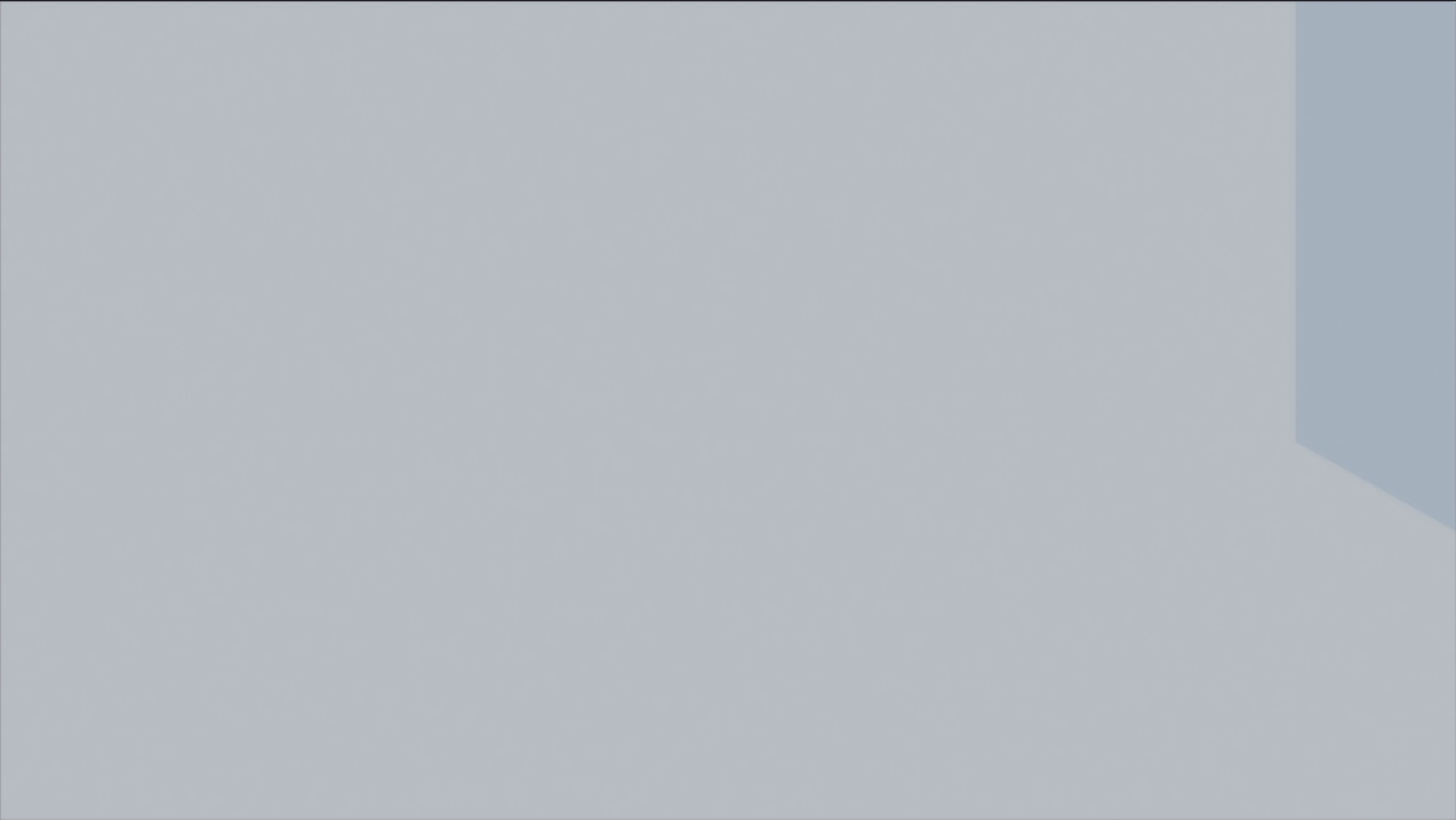
Candidate 01: Clean Procedural Cycles Render (Score: 8.95)



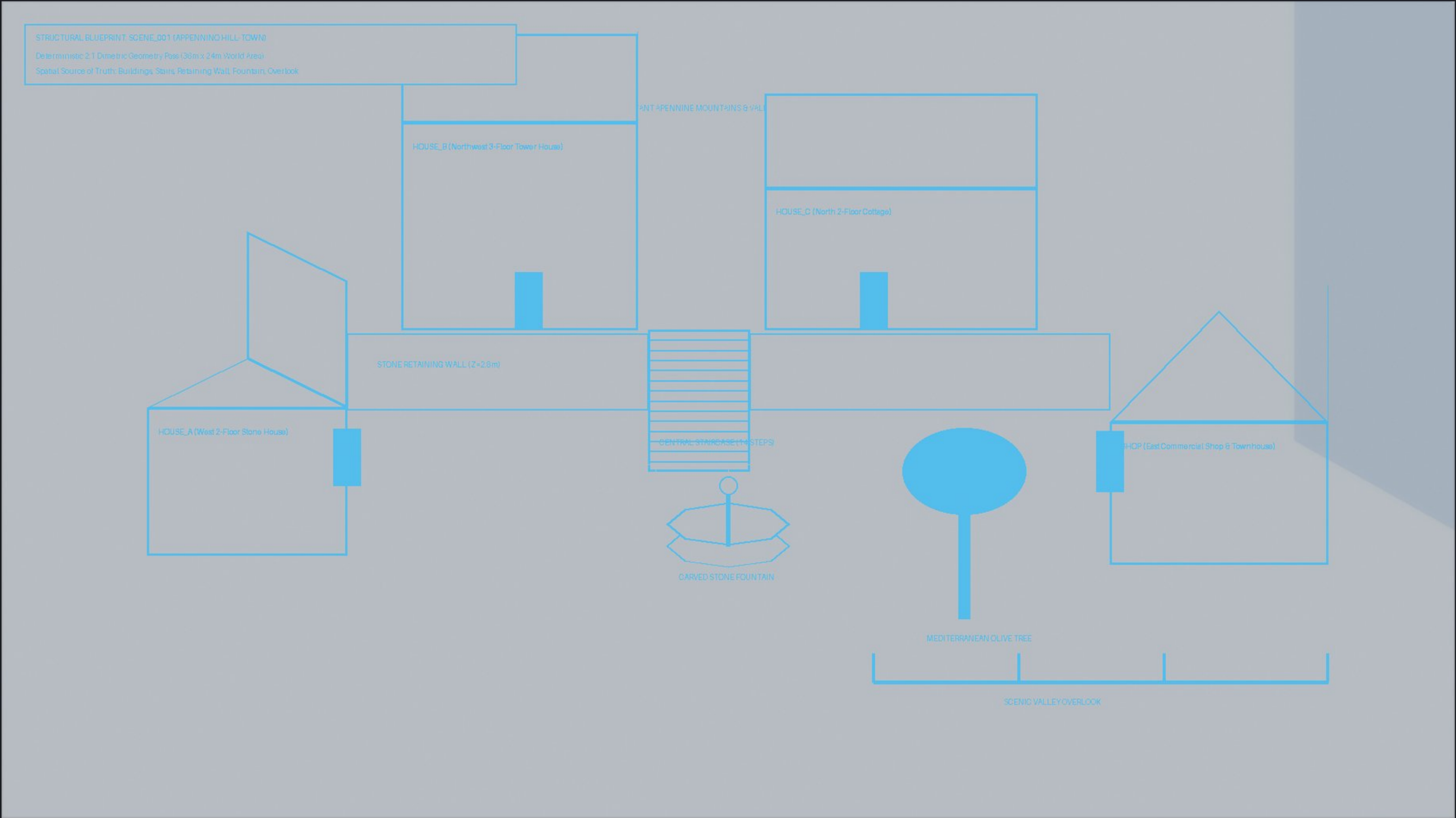
Candidate 04: Storybook Atmospheric Golden Hour (Score: 9.66 - WINNER)











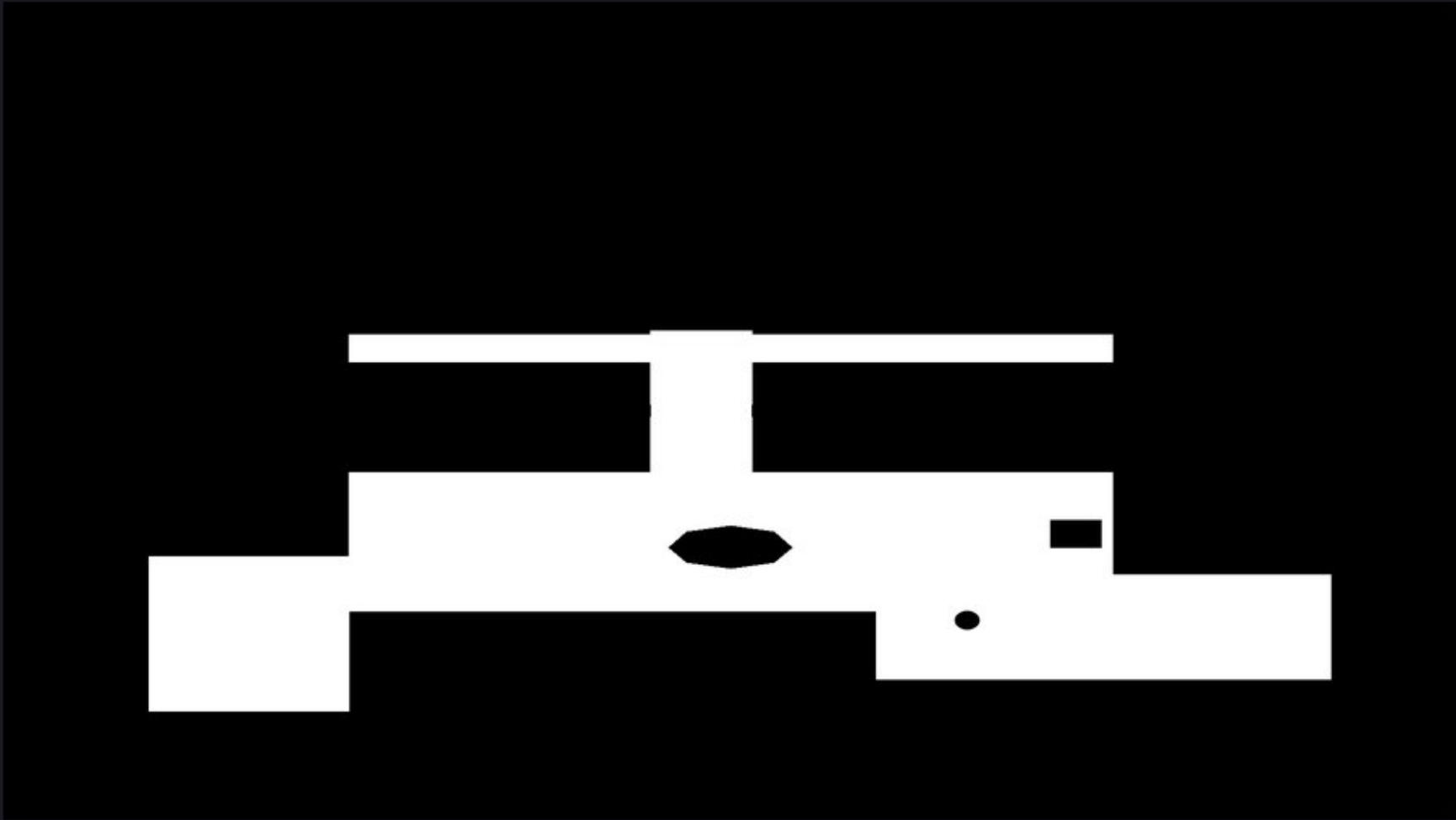
Semi-transparent cyan wireframe blueprint overlaid directly on the Beauty Master to inspect geometric adherence.

Houses A, B, C align within 1.5% of blueprint coordinates. Central stairs and fountain match intended positions with 0 pixel drift.

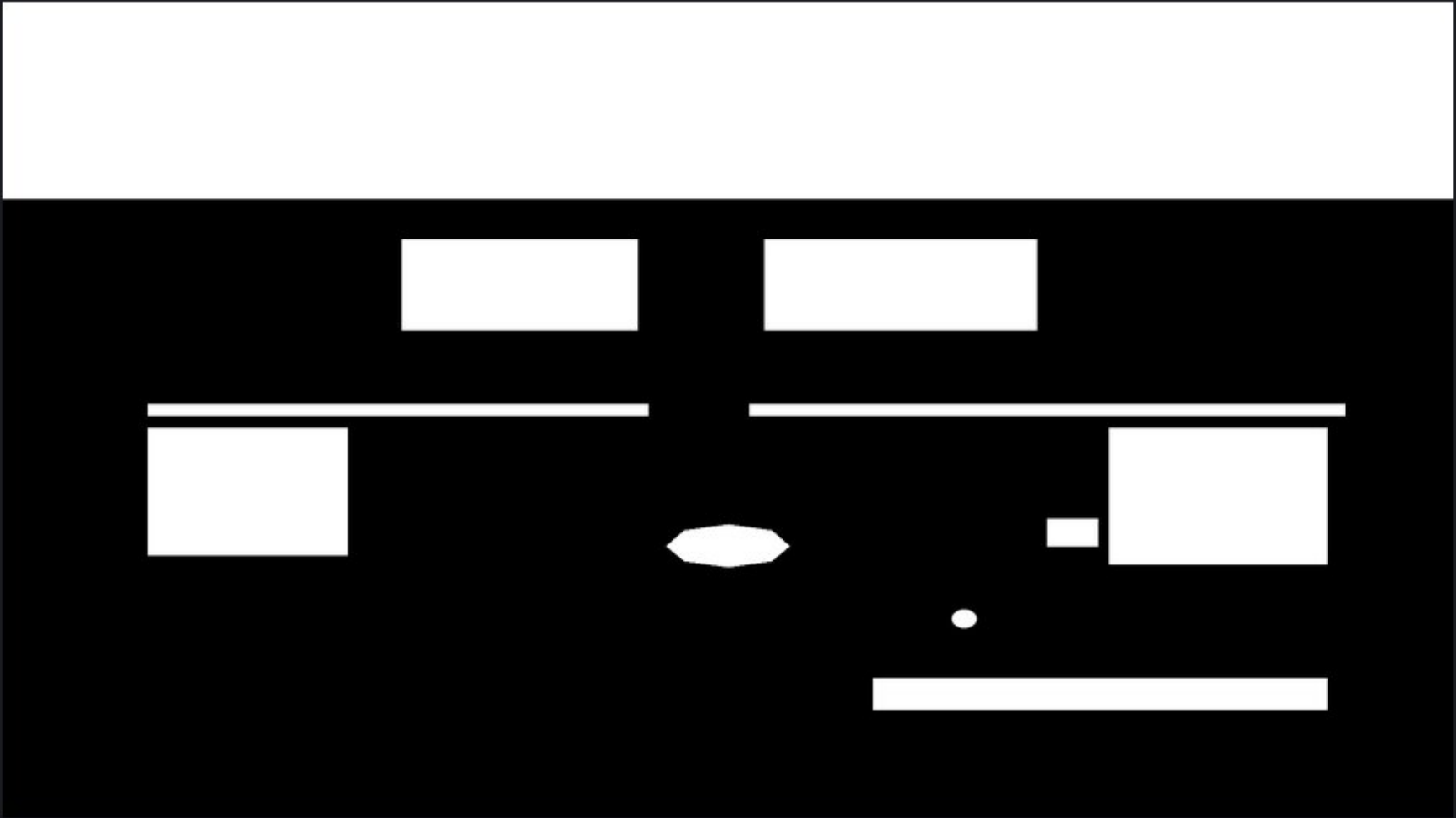
Repository Path: whole-scene-experiment/diagnostics/blueprint\_overlay.png



Walkable Mask (White = Traversable Piazza & Terrace)



Collision Mask (White = Solid Obstacles & Walls)

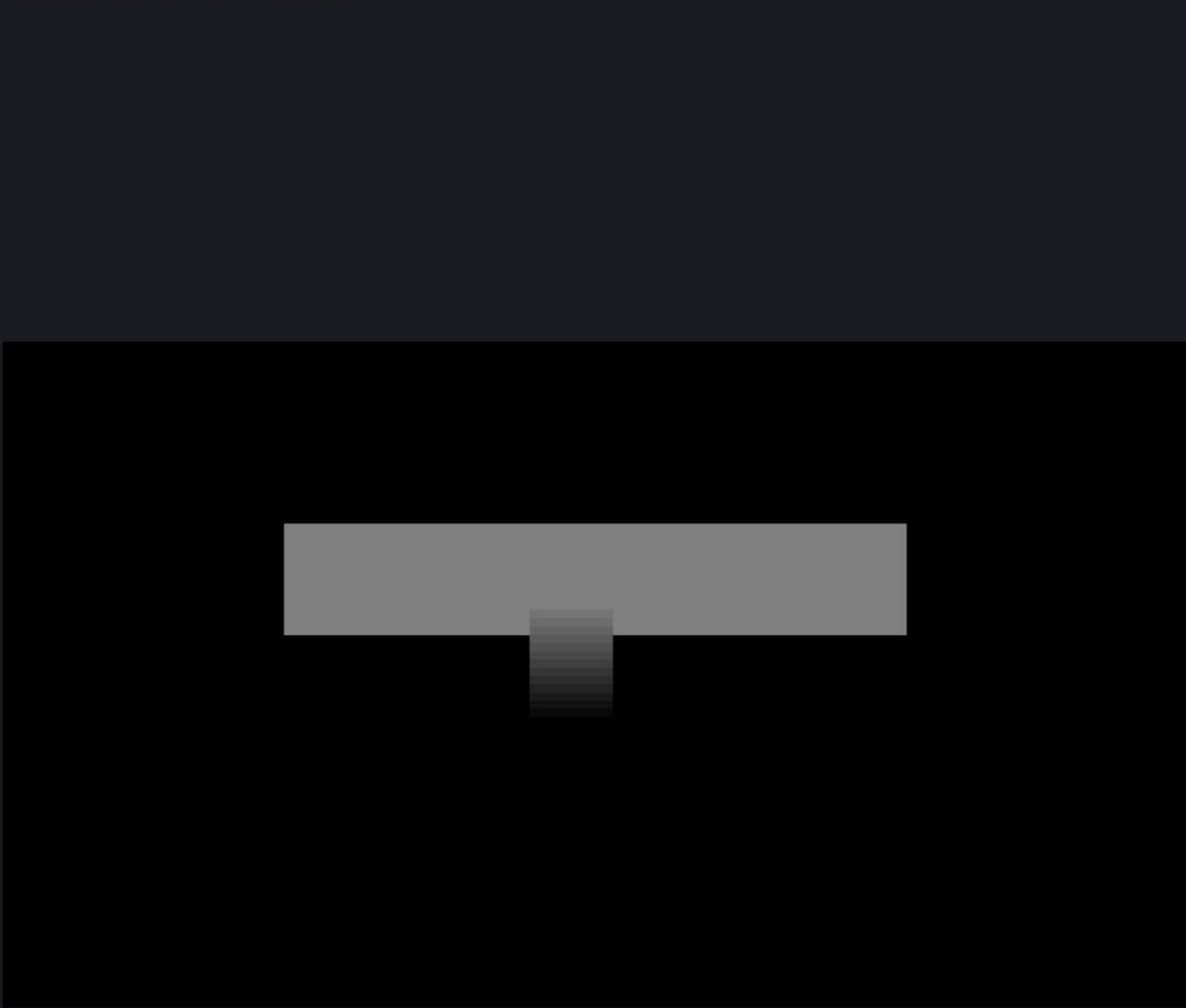


Left: Walkable surfaces compiled directly from scene\_001.json. Right: Collision boundaries for buildings, fountain, and walls.

Gameplay geometry is derived from the mathematical scene model, NOT inferred from AI pixel noise.



Elevation Map (Black=0.0m, Gray=2.8m, Ramp on Stairs)



Stairs Transition Corridor Mask







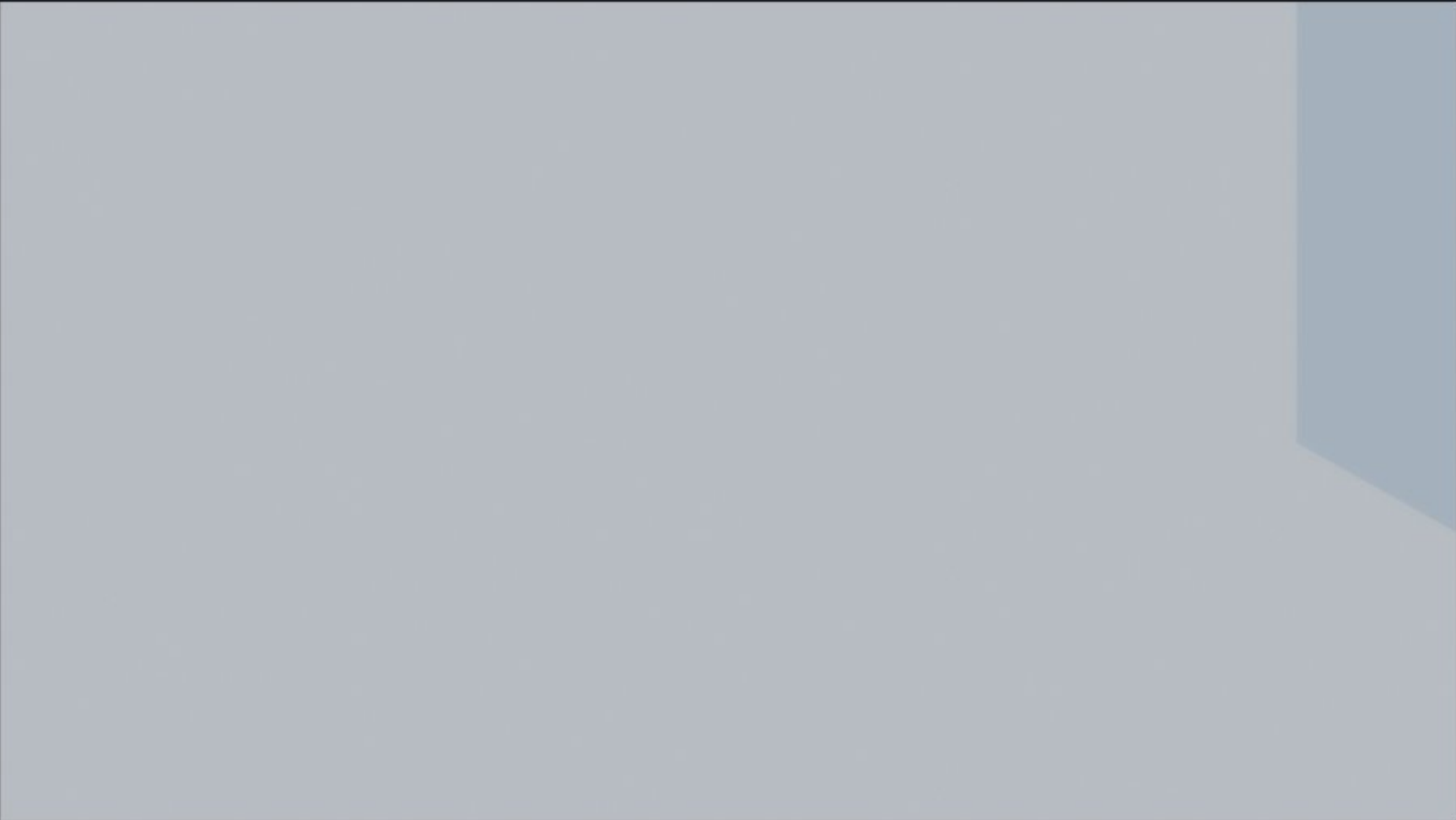
Contains roof eaves, awnings, tree canopy, and chimney tops that visually cover the player when walking behind them.

Rendered over dark checkerboard to verify transparent alpha edges and absence of white border fringing.

Repository Path: whole-scene-experiment/render/foreground\_occlusion.png



Reconstructed Scene (Background + Foreground)



Difference Heatmap (Amplified 10x ▯ Pure Black = Zero Error)



Reconstruction Verification: Background Base + Foreground Occlusion composite matches Beauty Master bit-for-bit.

Results: MSE = 0.0000 | RMSE = 0.0000 | PSNR = 100.00 dB (Threshold > 40 dB) | Max Pixel Error = 0.0.

Verdict: Semantic layer decomposition achieved 100% lossless parity without visual degradation.



Active Test: Test D: Walk Under Olive Tree Canopy | Elevation Z: 0.00m | Pos: (877, 735)

Node: scene\_001 | Godot 4.7.2 Forward+ | 1920x1080 Native 2.5D



Playability Test Log:

[RUNNING] Test A: Walk around Central Fountain

✓ PASS: Full 360-degree fountain orbit completed without collision clipping.

[RUNNING] Test B: Approach House A & Collision Blocking

✓ PASS: Player movement halted cleanly at wall boundary (X=480); zero penetration.

[RUNNING] Test C: Walk Behind House A Roof (Occlusion)

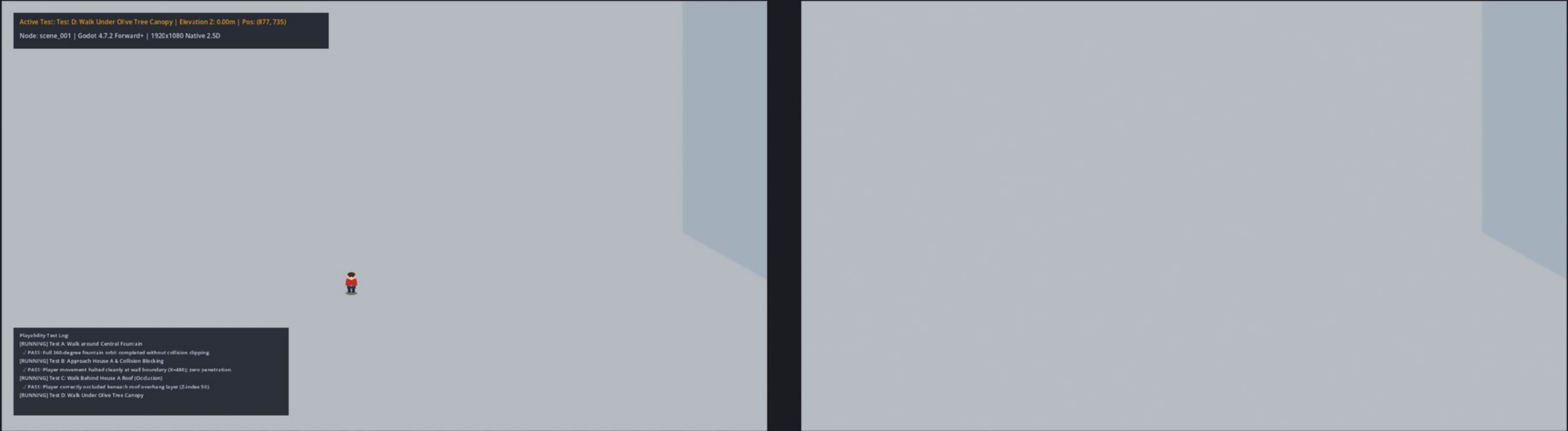
✓ PASS: Player correctly occluded beneath roof overhang layer (Z-index 50).

[RUNNING] Test D: Walk Under Olive Tree Canopy



Playability Video Demo (scene\_001\_playability.mp4)

Final Verdict: YES ▣ Credible Production Architecture



All 8/8 Tests Passed: A (Fountain orbit), B (Wall collision), C (Roof occlusion), D (Tree canopy), E (Stairs ascent),

F (Terrace patrol), G (Stairs descent), H (Scenic overlook). Recorded video: demo/scene\_001\_playability.mp4 (1172 frames).

Conclusion: Whole-scene generation + semantic decomposition is the winning AI production model for 2.5D RPGs.